

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech/Int.Btech Winter 2025 - 26 Examination

Semester: 07/11

Subject Code: 203105443

Subject Name: Big Data Analytics

Date: 27-12-2025

Time: 2:00pm to 4:30pm

Total Marks: 60

Instructions:

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1 is compulsory**, From Section B, **Q.1 is compulsory**.
3. Figures to the right indicate full marks.
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION-A		Marks	CO	BT
Q.1	Answer the following questions.			
	A. Explain: i. How NoSQL handles data differently ii. What are the Scalability differences iii. Flexibility in Big Data	[06]	CO1	BT1
	B. Compare SQL, NoSQL, NewSQL (differences, suitability for Big Data)	[06]	CO4	BT2
Q.2	A. What are the main features of Big Data?	[04]	CO1	BT1
	B. Demonstrate how HDFS supports high availability	[05]	CO3	BT3
	OR			
	B. How JSON can be used in Hadoop	[05]	CO5	BT3
Q.3	A. Define Hadoop and mention its primary purpose	[04]	CO3	BT4
	B. what are the Significance of MapReduce in Hive	[05]	CO5	BT2
	OR			
	B. Explain process of partitioning/combining in NoSQL	[05]	CO4	BT3
SECTION-B				
Q.1	Answer the following questions.			
	A. Explain: i. Apache Hive and its role ii. Hive integration with Hadoop iii. Advantages of Hive	[06]	CO5	BT4
	B. Elaborate: i. Role of MapReduce in Hive query execution ii. How it processes data iii. Advantages for Big Data analytics	[06]	CO5	BT2
Q.2	A. How NoSQL handles scalability and availability	[04]	CO4	BT2
	B. Machine learning integration with Big Data	[05]	CO1	BT3
	OR			
	B. Shuffle & Sort in MapReduce	[05]	CO3	BT4
Q.3	A. How Hadoop distributes tasks among nodes	[04]	CO3	BT3
	B. Importance of fault tolerance in Hadoop	[05]	CO3	BT2
	OR			
	B. How key-value stores ensure data integrity	[05]	CO4	BT4